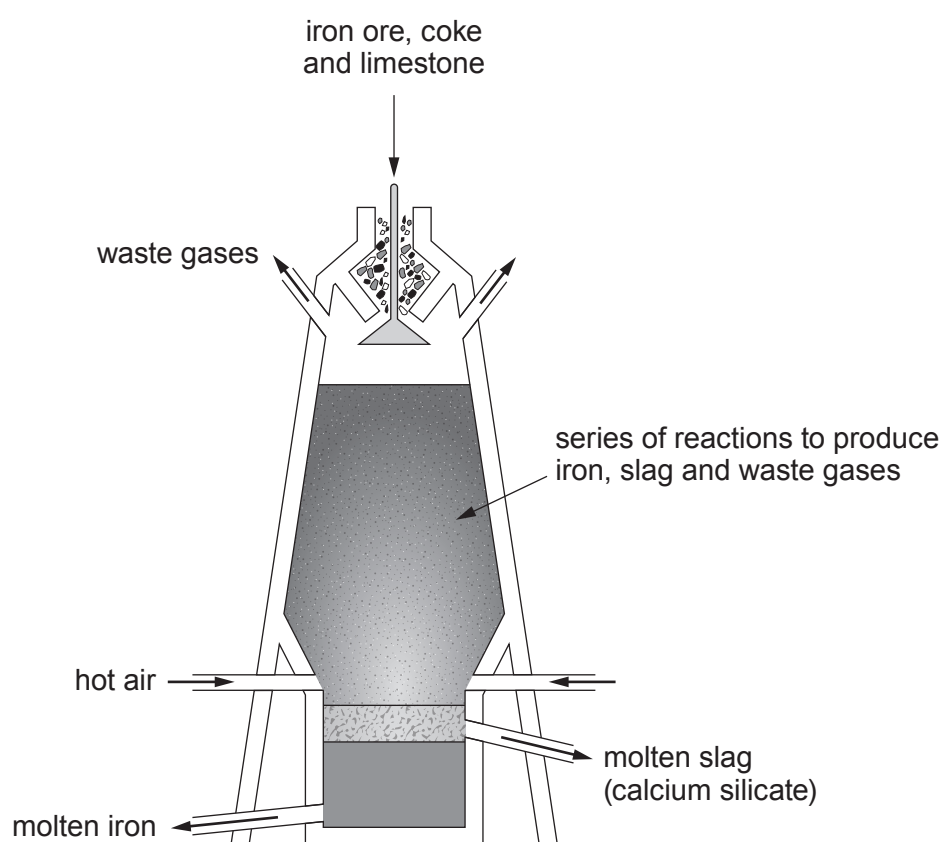


2. The diagram shows a blast furnace used to extract iron from iron ore.



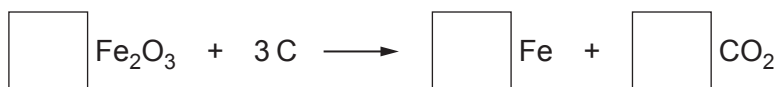
- (a) One of the reactions inside the furnace is the reaction between carbon dioxide and carbon, in the form of coke, to form carbon monoxide.

Give the symbol equation for this reaction.

[2]



- (b) This equation represents one of the reactions that produce iron inside the furnace.



- (i) Balance the equation. [1]

- (ii) Explain why this reaction shows that both oxidation and reduction take place.

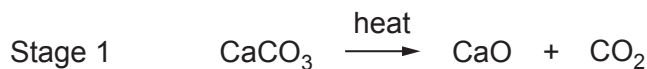
Refer to oxygen in your answer. [2]

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- (c) The formation of slag inside the furnace involves a two-stage process as summarised by the following equations.



- (i) Complete the name of the type of reaction taking place in stage 1. [1]

thermal

- (ii) Give the formula of calcium silicate, formed in stage 2. [1]

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